Data Science for Innovation

W4: Data Mining – Association Rule Mining

Presented by
Ali Anaissi

Overview of Week 4

Today: Data Mining– Association Rule Mining

Objective

Learn techniques for unsupervised learning, with tools in Python.

Lecture

- Association rule mining

Readings

- Intro to Data Mining, Ch. 6
  http://www-users.cs.umn.edu/~kumar/dmbook/ch6.pdf
- Intro to Data Mining, Ch. 8
  http://www-users.cs.umn.edu/~kumar/dmbook/ch8.pdf
- Data Science from Scratch, Ch. 11&19

Exercises

- Associations from scratch
- Associations using mlxtend
- Associations using pyfpgrowth

Supervised vs. Unsupervised Learning

- Supervised learning (e.g. classification and regression)
  - Supervision: The training data are accompanied by **labels** indicating the class of the observations
- Unsupervised learning (e.g. clustering and association rules)
  - The class labels of training data is **unknown**
  - Given a set of measurements, observations, etc. with the aim of
    - Establishing the existence of classes or clusters in the data
    - Discovering hidden patterns or rules

Unsupervised Learning

- We'll focus on unsupervised machine learning techniques
  - *Association rule mining*
  - Dimensionality reduction
  - Outlier detection
  - *Clustering*
  - Etc.

Association Rule Mining

Association Analysis

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Market-basket transactions

TID: Transaction Identifier

Items: Transaction item set

How can businesses improve sales by analysing customer purchase data?

Slides adapted from Tan et al. Introduction to data mining.
http://www-users.cs.umn.edu/~kumar/dmbook/
http://www-users.cs.umn.edu/~kumar/dmbook/dmslides/chap6_basic_association_analysis.pdf

Association Rule Mining

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Market-basket transactions

TID: Transaction Identifier

Items: Transaction item set

– Predict occurrence of an item based on other items in the transaction, eg:

{Diaper} → {Beer}
{Milk,Bread} → {Eggs,Coke}
{Beer,Bread} → {Milk}

– Note that arrows indicate co-occurrence, not causality

Definition: Itemset

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Market-basket transactions

TID: Transaction Identifier

Items: Transaction item set

– An **itemset** is a collection of one or more items

  {Milk,Bread,Diaper}

– A **k-itemset** is an itemset containing k items

Definition: Frequent Itemset

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Market-basket transactions

TID: Transaction Identifier

Items: Transaction item set

- **Support count** (σ) is the itemset frequency

  $$\sigma(\{\text{Milk}, \text{Diaper}, \text{Beer}\}) = 2$$

- **Support** (s) is the normalised itemset frequency

  $$s = \frac{\sigma(\{\text{Milk}, \text{Diaper}, \text{Beer}\})}{|T|} = \frac{2}{5}$$

- A **frequent itemset** has $s \geq min\_support$

Definition: Association Rule

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Market-basket transactions

TID: Transaction Identifier

Items: Transaction item set

- An **association rule** is an implication of the form X→Y where X and Y are itemsets
  {Milk,Diaper}→{Beer}
- **Confidence** (c) measures how often Y occurs in transactions with X

$$c = \frac{\sigma(\{\text{Milk,Diaper,Beer}\})}{\sigma(\{\text{Milk,Diaper}\})} = \frac{2}{3}$$

- An **association rule** has $c \geq$ min_conf

Mining Association Rules

1. **Frequent itemset generation**
   – Generate all itemsets with $s \geq min\_support$
2. **Rule generation**
   – Generate high-confidence rules from each frequent itemset
   – Each rule is a binary partitioning of a frequent itemset

Finding Frequent Itemsets

TID	Items
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

Market-basket transactions

TID: Transaction Identifier

Items: Transaction item set

- Let *min_support = 50%*
- Freq. 1-itemsets:
  - Beer:3(60%); Nuts:3(60%); Diaper:4(80%); Eggs:3(60%)
- Freq. 2-itemsets:
  - {Beer, Diaper}:3(60%)

Finding Association Rules

TID	Items
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

Market-basket transactions

TID: Transaction Identifier

Items: Transaction item set

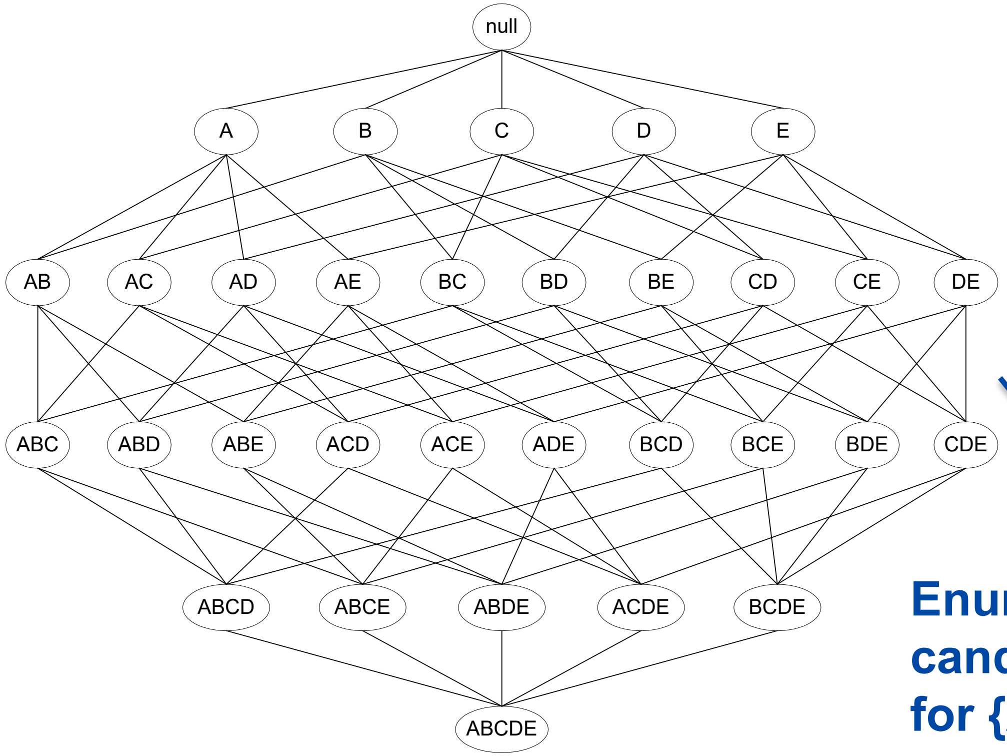
- Let *min_support* = 50%, min_conf = 50%
- Freq. Pat.: Beer:3, Nuts:3, Diaper:4, Eggs:3, {Beer, Diaper}:3
- Association rules:
  - Beer → Diaper (100%)
  - Diaper → Beer (75%)

Mining Association Rules

1. **Frequent itemset generation**
   – Generate all itemsets with $s \geq min\_support$
2. **Rule generation**
   – Generate high-confidence rules from each frequent itemset
   – Each rule is a binary partitioning of a frequent itemset

Easy! But brute force enumerate is computationally prohibitive..

There are 2^d candidate itemsets!



Enumeration of 2^5-1 candidate itemsets for {A,B,C,D,E}

Reducing the number of candidates

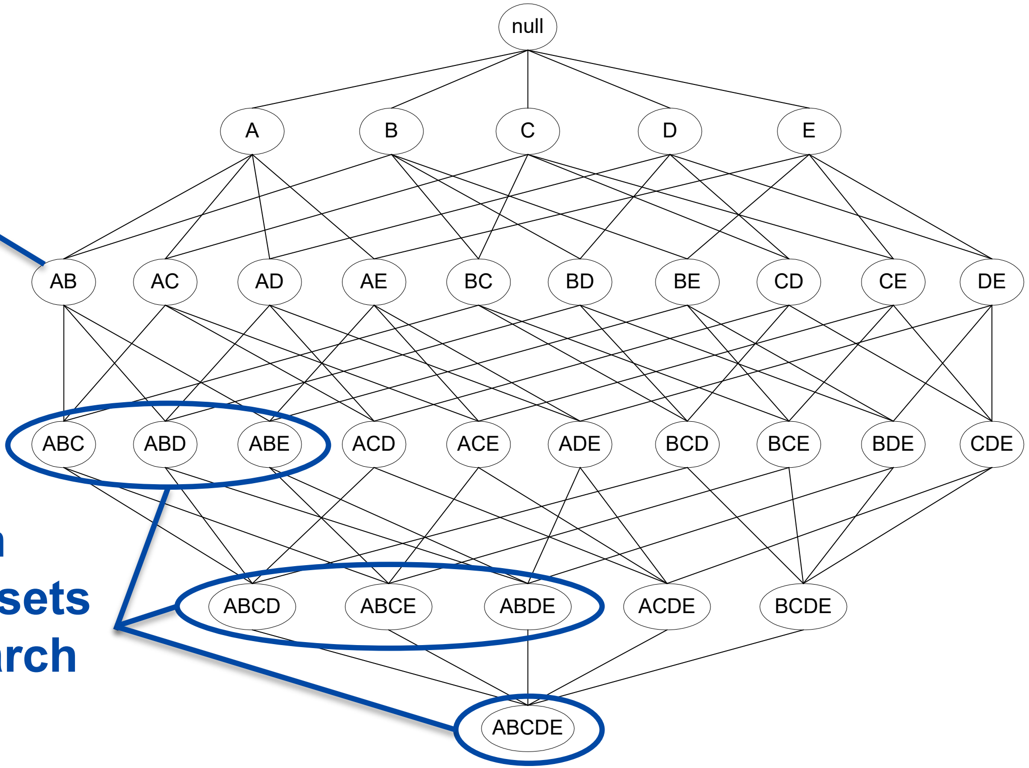
- **Apriori Principle**
  If an itemset is frequent, then all of its subsets are also frequent

- **Conversely**
  If an itemset is infrequent, then its supersets are also infrequent
  - The support of an itemset never exceeds the support of its subsets
    - s (Bread) $\geq$ s (Bread, Beer)
    - s (Milk) $\geq$ s (Bread, Milk)
  - This is known as the anti-monotone property of support

Pruning the 2^d candidate itemsets

If *s({A,B})* ≤ *min_support*

Then we can prune supersets from the search space



Apriori algorithm for generating frequent itemsets

While the list of $(k\text{-}1)$-itemsets is non-empty:

  Generate candidate k-itemsets

  Identify and keep frequent k-itemsets

The Apriori Algorithm—An Example

Min_sup = 2

Database TDB

Tid	Items
1	A, C, D
2	B, C, E
3	A, B, C, E
4	B, E

1st scan → C_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

→ L_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{E}	3

C_2

Itemset
{A, B}
{A, C}
{A, E}
{B, C}
{B, E}
{C, E}

2nd scan → C_2

Itemset	sup
{A, B}	1
{A, C}	2
{A, E}	1
{B, C}	2
{B, E}	3
{C, E}	2

→ L_2

Itemset	sup
{A, C}	2
{B, C}	2
{B, E}	3
{C, E}	2

C_3

Itemset	sup
{B, C, E}	2

3rd scan → L_3

Itemset	sup
{B, C, E}	2

Create initial 1-itemsets

Add each item to the initial list of candidate itemsets

```
def createC1(dataset):
    "Create a list of candidate item sets of size one."
    c1 = []
    for transaction in dataset:
        for item in transaction:
            if not [item] in c1:
                c1.append([item])
    c1.sort()
    #frozenset because it will be a ket of a dictionary.
    return list(map(frozenset, c1))
```

Sort and return as list of sets

Identify itemsets that meet the support threshold

```
def scanD(dataset, candidates, min_support):
    "Returns all candidates that meets a minimum support level"
    sscnt = {}
    for tid in dataset:
        for can in candidates:
            if can.issubset(tid):
                sscnt.setdefault(can, 0)
                sscnt[can] += 1

    num_items = float(len(dataset))
    retlist = []
    support_data = {}
    for key in sscnt:
        support = sscnt[key] / num_items
        if support >= min_support:
            retlist.insert(0, key)
        support_data[key] = support
    return retlist, support_data
```

Calculate support counts for each candidate

Check whether candidates meet threshold

Generate the next list of candidates

(k-1)-itemsets

Iterate through all pairs of itemsets

```python
def aprioriGen(freq_sets, k):
    "Generate the joint transactions from candidate sets"
    retList = []
    lenLk = len(freq_sets)
    for i in range(lenLk):
        for j in range(i + 1, lenLk):
            L1 = list(freq_sets[i])[:k - 2]
            L2 = list(freq_sets[j])[:k - 2]
            L1.sort()
            L2.sort()
            if L1 == L2:
                retList.append(freq_sets[i] | freq_sets[j]) # | is set union
    return retList
```

Check whether pairs differ by a single item

A|B returns the union of A and B

Generate all Frequent Itemsets

Initialise L with frequent 1-itemsets

```
def apriori(dataset, min_support=0.5):
    "Generate a list of candidate item sets"
    C1 = createC1(dataset)
    D = list(map(set, dataset))
    L1, support_data = scanD(D, C1, min_support)
    L = [L1]
    k = 2
    while (len(L[k - 2]) > 0):
        Ck = aprioriGen(L[k - 2], k)
        Lk, supK = scanD(D, Ck, min_support)
        support_data.update(supK)
        L.append(Lk)
        k += 1

    return L, support_data
```

While the list of (k-1)-itemsets is non-empty:

Generate candidate k-itemsets

Identify frequent k-itemsets

Keep frequent k-itemsets

Exercise: Generating Frequent Itemsets

- Frequent itemset generation
  - code cell after “Generate frequent itemsets”
  - code cell after “Itemset generation on sample data”
- Exploring support
  - Try different support thresholds
  - What’s a reasonable value for real data?
  - Do you have datasets that resemble transactions?
  - What about the apps/websites you use?

Rule Generation

- Given a frequent itemset **L**, find all non-empty subsets $\mathbf{x} \subset \mathbf{L}$ such that $\mathbf{x} \rightarrow \mathbf{L} - \mathbf{x}$ satisfies the minimum confidence requirement
  - If {A,B,C,D} is a frequent itemset, candidate rules:

    ABC →D, ABD →C, ACD →B, BCD →A,

    A →BCD, B →ACD, C →ABD, D →ABC,

    AB →CD, AC → BD, AD → BC, BC →AD,

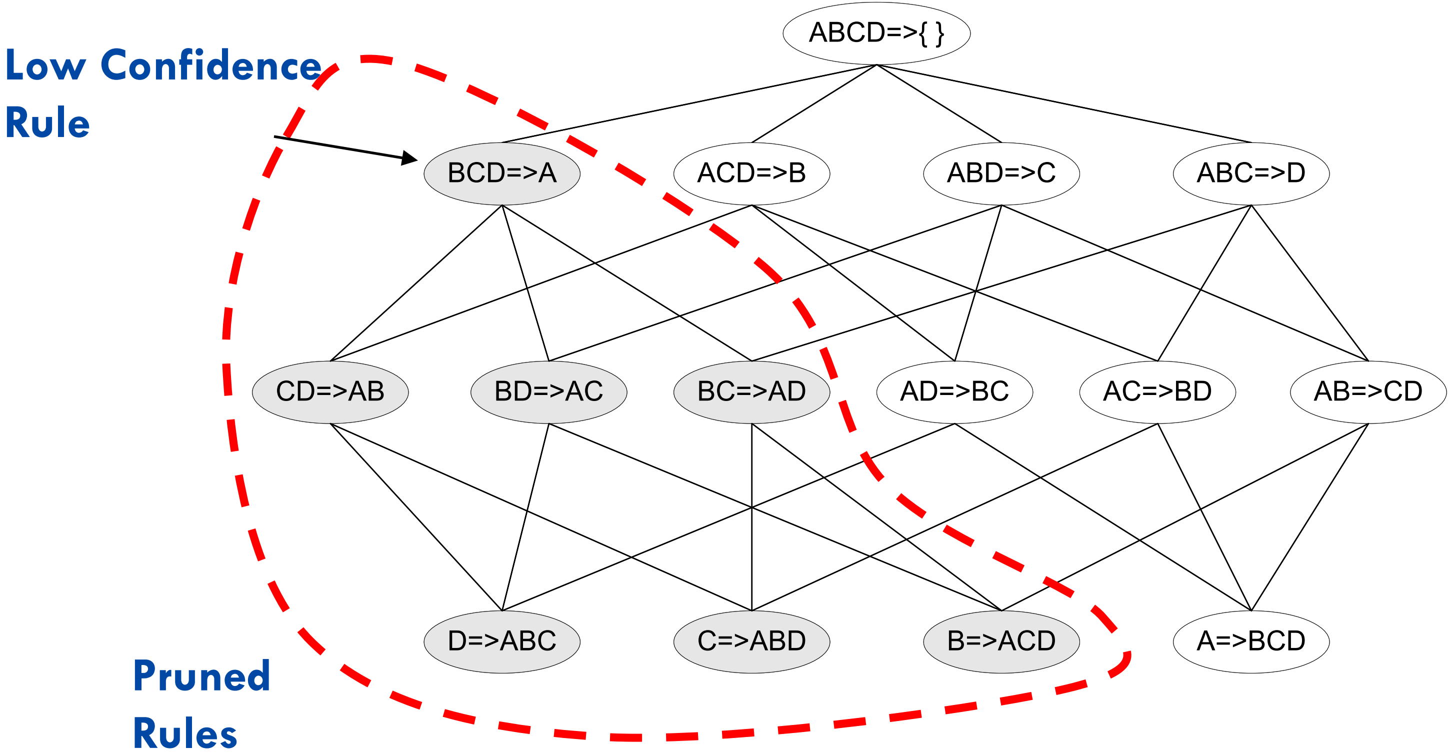
    BD →AC, CD →AB,

- If $|\mathbf{L}| = \mathbf{k}$, then there are $\mathbf{2^k - 2}$ candidate association rules

Efficient Rule generation

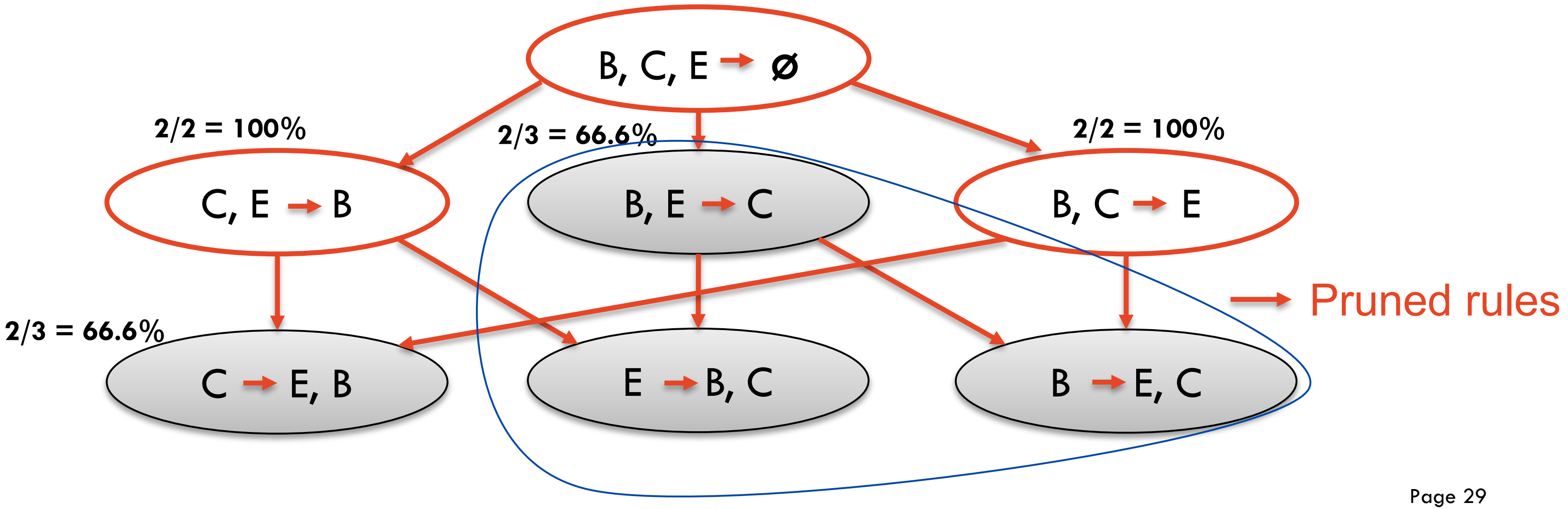
- How to efficiently generate rules from frequent itemsets?
- Similar to support, the confidence of rules generated from the same itemset has an anti-monotone property
- Example: X = {A,B,C,D}:
  - $c(ABC \rightarrow D) \geq c(AB \rightarrow CD) \geq c(A \rightarrow BCD)$
- Confidence is anti-monotone w.r.t. number of items on the RHS of the rule

Rule Generation for Apriori Algorithm



Association Rule

- From the example before, we had a frequent itemset: {B, C, E}
- From this frequent itemset {B, C, E}, we can have the following association rules



Exercise: Association Rule Mining

- Frequent itemset generation
  - code cell after “Mine association rules”
  - code cell after “Rule mining on sample data”
- Exploring confidence
  - Try different confidence thresholds
  - What’s a reasonable value for real data?
  - Can we use this for recommendation (e.g., Amazon, Netflix)?
- Using mlxtend library
  - code cell after “Association analysis using mlxtend library”
  - code cell after “Load the supermarket transaction datasets”
  - Mine association rules on Groceries datasets

Association Rule Mining with FP-Growth

Performance Bottlenecks of Apriori

- Bottlenecks of *Apriori*:
  - Candidate generation:
    - Generate huge candidate sets
  - Multiple scans of database

Overview of FP-Growth:

- Compress a large database into a compact, *Frequent-Pattern tree* (FP-tree) structure
  - Highly compacted, but complete for frequent pattern mining
  - Avoid costly repeated database scans
- Develop an efficient, FP-tree-based frequent pattern mining method (FP-growth)
  - A divide-and-conquer methodology: decompose mining tasks into smaller ones
  - Avoid candidate generation

Construct FP-tree

Two Steps:

1. Scan the transaction DB for the first time, find frequent items (single item patterns) and order them into a list L in frequency descending order.

2. For each transaction, order its frequent items according to the order in L; Scan DB the second time, construct FP-tree by putting each frequency ordered transaction onto it.

FP-tree Example: step 1

Step 1: Scan DB for the first time to generate L

Min_sup = 2

TID	Items bought
1	A, C, D
2	B, C, E
3	A, B, C, E
4	B, E

Item	Frequency
A	2
B	3
C	3
D	1
E	3

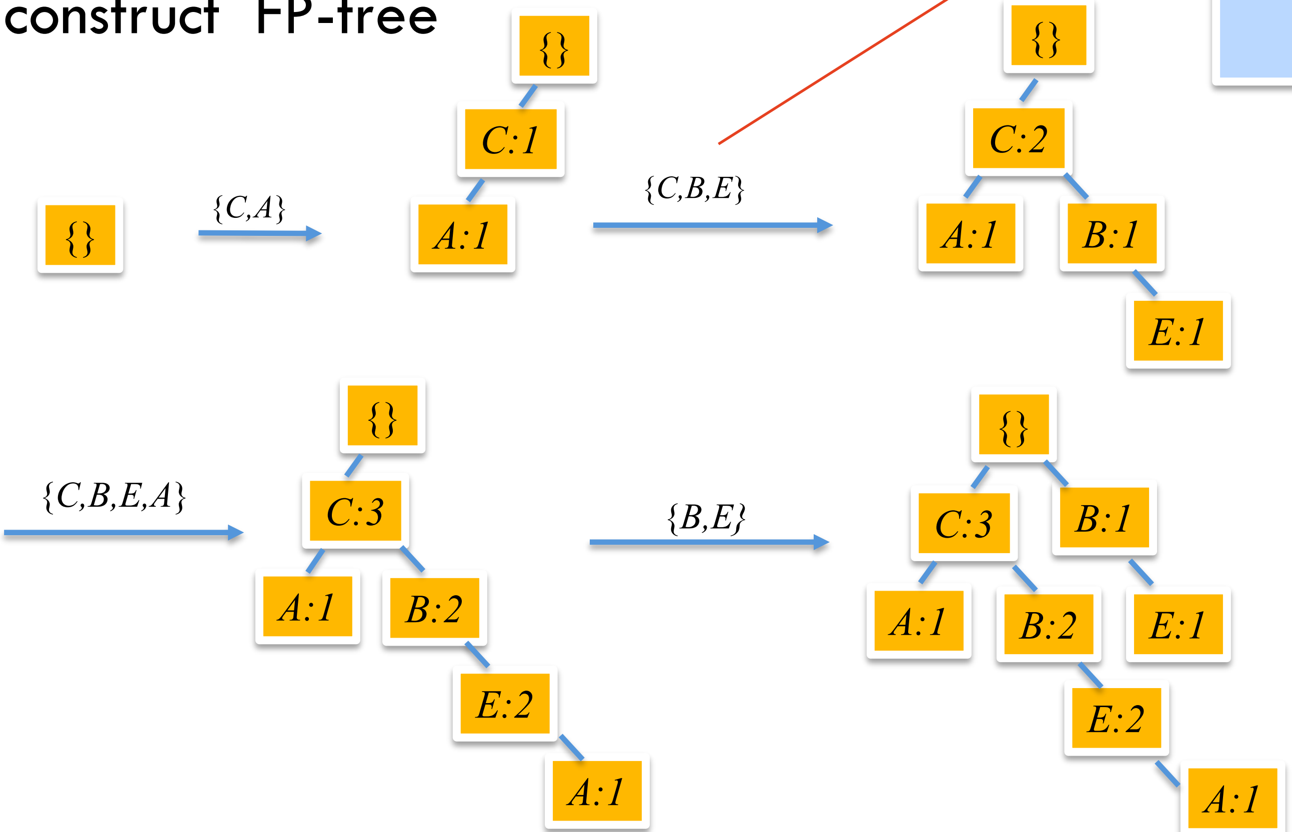
FP-tree Example: step 2

Step 2: scan the DB for the second time, order frequent items in each transaction

TID	Items bought	Ordered frequent items
1	A, C, D	C, A
2	B, C, E	C, B, E
3	A, B, C, E	C, B, E, A
4	B, E	B, E

FP-tree Example: step 2

Step 2: construct FP-tree



NOTE: Each transaction corresponds to one path in the FP-tree

Mining Frequent Patterns Using FP-tree

Starting the processing from the end of list L:

Step 1:

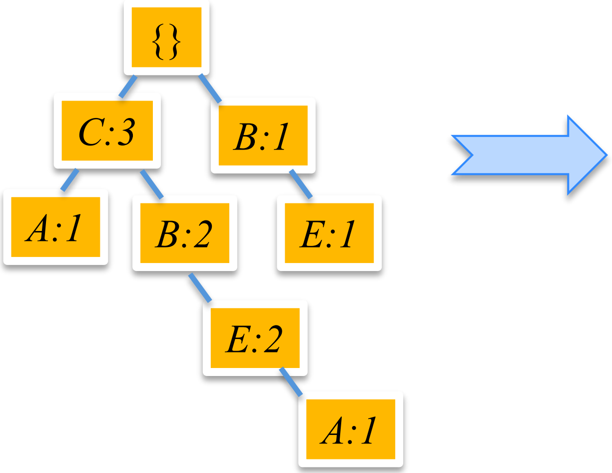
Construct **conditional pattern base** for each item in the header table

Step 2:

Construct **conditional FP-tree** from each conditional pattern base

Step 1: Construct Conditional Pattern Base

- Starting at the bottom of frequent-item header table in the FP-tree
- Traverse the FP-tree by following the link of each frequent item
- Accumulate all of **transformed prefix paths** of that item to form a **conditional pattern base**



***Conditional* pattern bases**

Item	Cond. Pattern base
A	EBC:1, C:1
E	BC:2, B:1
B	C:2, {}
C	{}

Step 2: Construct Conditional FP-tree

- For each pattern base
  - Accumulate the count for each item in the base
  - Construct the conditional FP-tree for the frequent items of the pattern base

A-cond. **pattern base:**		*A-conditional* **FP-tree**	**Frequent patterns relate to *A*:**
EBC:1, C:1	→	**{ (C:2) } | A**	**AC:2**

Step 2: Construct Conditional FP-tree

***E**-cond.* **pattern base:**

BC:2, B:1

→ ***E**-conditional* **FP-tree**

{ (B:3, C:2, BC:2) } | E

Frequent patterns relate to *E*:

EB :3, EC:2, BCE:2

***B**-cond.* **pattern base:**

C:2

→ ***B**-conditional* **FP-tree**

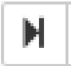
{ (C:2) } | B

Frequent patterns relate to *B*:

BC :2

All Frequent Item sets: { A:2, E:3, B:3, C:3,
BC:2, EC:2, AC:2, EB:3,
BCE:2 }

Exercise: Association Rule Mining Using FP-Growth

- Frequent itemset generation
  -  code cell after “Rules generation”
- Mine association rules using FP-growth on Groceries datasets
  - Try different confidence thresholds
  - What’s a reasonable value for real data?

Review

Today: Data Mining – Association Rule Mining

Objective

Learn techniques for unsupervised learning, with tools in Python.

Lecture

– Association rule mining

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– Intro to Data Mining, Ch. 8
http://www-users.cs.umn.edu/~kumar/dmbook/ch8.pdf

– Data Science from Scratch, Ch. 11&19

Exercises

– Associations from scratch

– Associations using mlxtend

– Associations using pyfpgrowth

TODO in W5

– define experimental framework for A2

Additional Reading (not examinable)

- Tan et al. Introduction to data mining.
  https://goo.gl/hWwuZb
- Aggarwal. Data mining: the textbook.
  https://goo.gl/lQqLwT
- Han. Data mining: concepts and techniques.
  https://goo.gl/CFIMMs
- Scikit-learn user guide, § 2 (Unsupervised learning).
  http://scikit-learn.org/stable/unsupervised_learning.html

Other tools and Techniques (not examinable)

- Scikit-learn user guide, § 4.4 (Dimensionality reduction).
  http://scikit-learn.org/stable/modules/unsupervised_reduction.html
- Scikit-learn user guide, § 2.7 (Outlier detection).
  http://scikit-learn.org/stable/modules/outlier_detection.html
- Etc

Project and discussion time

Time for you to talk

to tutors, instructors and each other.